

GLOW AUTOMATION

→ Defination:-

"The bright light of different colour emitted by a substance is control automatically this process is called glow automation. This process is completely automated."

The glow is a plasma formed by the passage of electric current through a low pressure gas. It is created by applying a voltage between two metal electrodes in a glass tube containing gas. When voltage exceeds a certain value called "striking voltage". The gas in the tube ionizes, becoming a plasma and begins conducting electricity, causing it to glow with a colour light. Colour depend on the gas used.

→ Glow Source:-

A low pressure glow source operating in the abnormal mode was used. The source has two vacuum ports. The lateral one act as the main exhaust port for the glass chamber, both

for initial evacuation and for flow operation. The upper part has a brass tubing of 0.15cm. This tubing serve as anode. The Cathode is replaceable Copper or graphite hemicylinder of 1.80cm length.

→ Basic Operating Mechanism:-

It consist of two electrodes in a cell held at low pressure (0.1-10⁻⁴ about 1/10000th to 1/100th of atmospheric pressure). The cell is filled with neon, but other gases can also be used. An electric potential of several volts is applied between two electrodes. The ions⁽⁺⁾ are driven toward the Cathode⁽⁻⁾ by electric potential and electron⁽⁻⁾ are driven toward anode by the same potential. The initial population of ions and electrons collides with other atoms, ionizing them. As long as potential is maintain.

Some of the ions, Kinetic energy is transferred to the Cathode. This happens partially through the ions striking the Cathode directly. The primary mechanism is less direct the ion collides with gas atom and transfer some energy and then atom strike the Cathode. The atom can be excited by collision with ions, electrons or other atoms that have been

previously excited by collisions. Once excited atom will lose their energy quickly. The wavelength of photon can be used to determine the identity of the atom and number of photon is directly proportion to the concentration of element in sample. Some collision will cause ionization.

→ Application:-

- 1- It is used in analytical chemistry.
- 2- It is used for the analysis of element, molecules and composition of solid, liquid and gases.
- 3- The wavelength of emitted light can be identify the substance.
- 4- Both bulk and depth analysis of solid may be performed with this method.
- 5- It is used to produce coloured light.